SPIRE ANALYTICS

How to run the paper-trading bot

Version generated 01 October 2026 01:27 (Skopje) from the bot's code and config.yaml. If something here disagrees with the app, regenerate this document (see the end).

1. What the bot is, and what it isn’t

- It trades a **paper account** at Alpaca: pretend money, real market prices. The code is hard-wired to paper trading; there is no switch to real money.
- It follows rules you choose (the **jobs**): each job is one strategy on one or more symbols with a share of the account. It checks the market once per 5Min bar and sends market orders.
- It protects itself: if the account is down 2.0% in a day it sells everything and stops for the day; no single symbol gets more than $10,000; stock positions are closed 5 minutes before the US close.
- It writes down why it acted (logs/decisions.jsonl) and produces a plain-English report after every trading day.

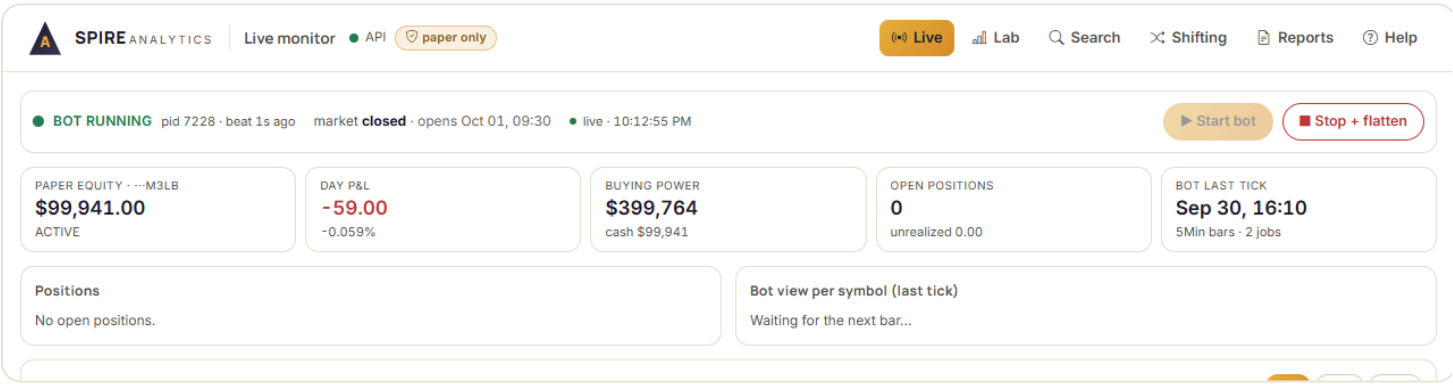
What it can't do

- It can't predict the market or guarantee a profit. Every strategy loses in some markets.
- It doesn't read the news; it only sees prices.
- It can't trade while the PC is off, asleep or offline, and it can't fix a broken internet connection.
- Market orders fill at the best available price, which can differ from the last price it saw, especially at the open and in fast markets.
- It never trades real money, and results on paper can be better than real trading would be.

2. Before the market opens

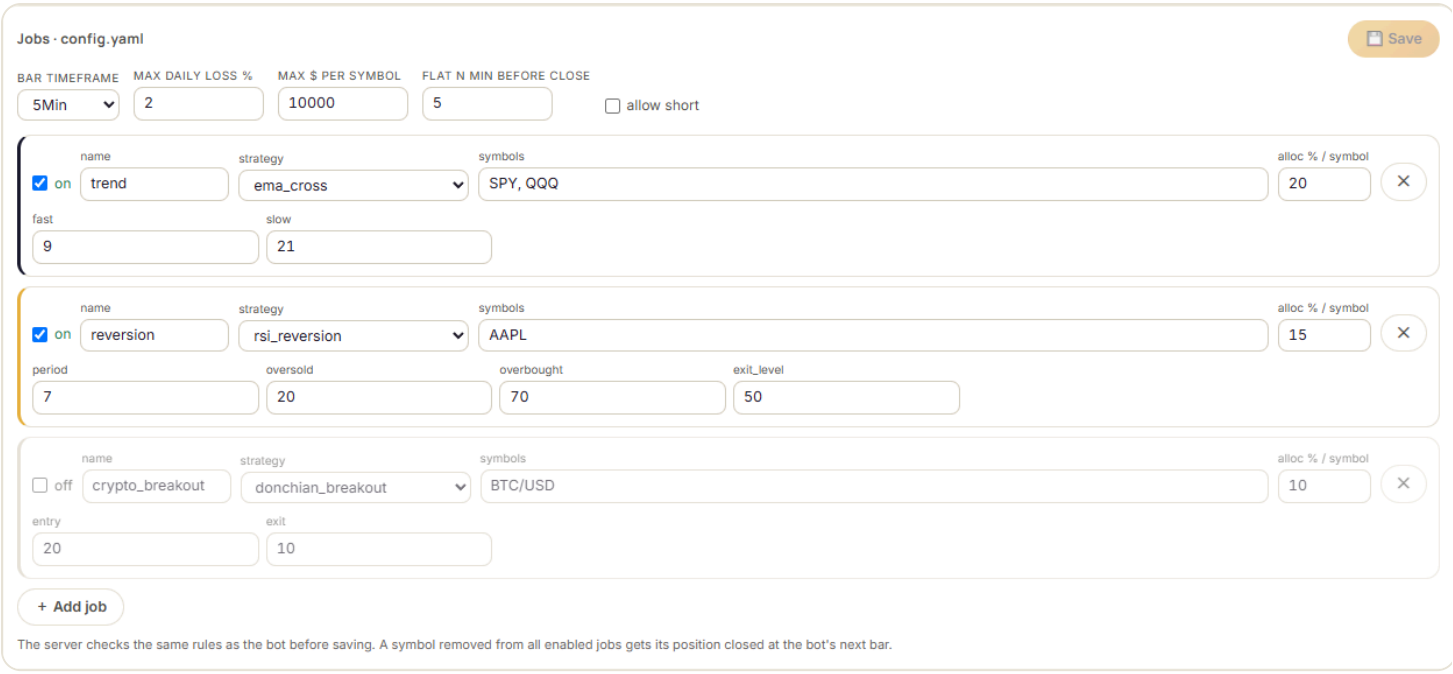
The US market opens at 09:30 New York time, which is usually **15:30** in Skopje (see the table in section 4 for the weeks when it differs).

1. Switch the PC on and make sure it won't sleep: Windows Settings → System → Power → *Sleep* = **Never** (when plugged in). The bot stops trading if the PC sleeps.
2. Double-click **start_lab.bat** in E:\autobot. Two windows open ("autobot API" and "autobot web") and the lab opens in the browser at http://127.0.0.1:5190. Leave both windows open all day.
3. Open the **Live** page. Check the small dot next to *live* is **green** (connected) and the header says **API** with a green dot and **paper only**.
4. Check *Paper equity* shows a dollar amount and the account status is ACTIVE.
5. Scroll to **Jobs · config.yaml** and check which jobs are switched **on** and on which symbols (below). Change nothing on the day unless it was tested in the Lab first.



The Live page when everything is connected: green dots next to API and live, the paper-only badge, the account figures.

JOB	ON?	STRATEGY	SYMBOLS	SHARE OF ACCOUNT PER SYMBOL
trend	on	ema_cross (follow the trend)	SPY, QQQ	20%
reversion	on	rsi_reversion (buy the dip)	AAPL	15%
crypto_breakout	off	donchian_breakout (buy the breakout)	BTC/USD	10%



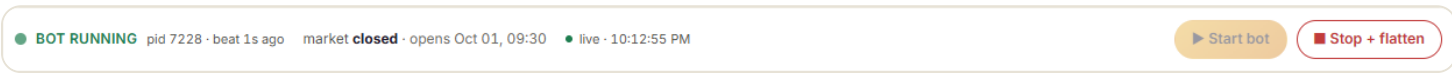
The Jobs editor. Switch a job on or off with its checkbox, then Save. The bot applies saved changes at its next bar.

Automatic start (optional)

The bot and the lab can start by themselves 2 minutes after you log on to Windows (Task Scheduler tasks **Spire autobot** and **Spire autobot lab**). Turn it on with **autostart_on.bat** and off with **autostart_off.bat** in E:\autobot; turning it off doesn't stop anything already running. Started this way, a supervisor restarts the bot after a crash (at most 3 times an hour) and writes each crash to logs/bot.log. If a bot is already running, a second one exits quietly.

3. Starting the bot

1. On the **Live** page press ▶ **Start bot** (top right), or double-click **start_bot.bat** (this one also restarts the bot after a crash).
2. Within a few seconds the status changes to **BOT RUNNING** with a green dot, a process number (pid) and *beat 1s ago*. The beat number must stay small (under 10 seconds).
3. Wait for the first tick: within one 5Min bar, *Bot last tick* shows the current time and the *Bot view per symbol* table fills in.
4. If Start is greyed out, a bot is already running (or a stuck one is still alive, shown as *unresponsive*). Never start a second bot.

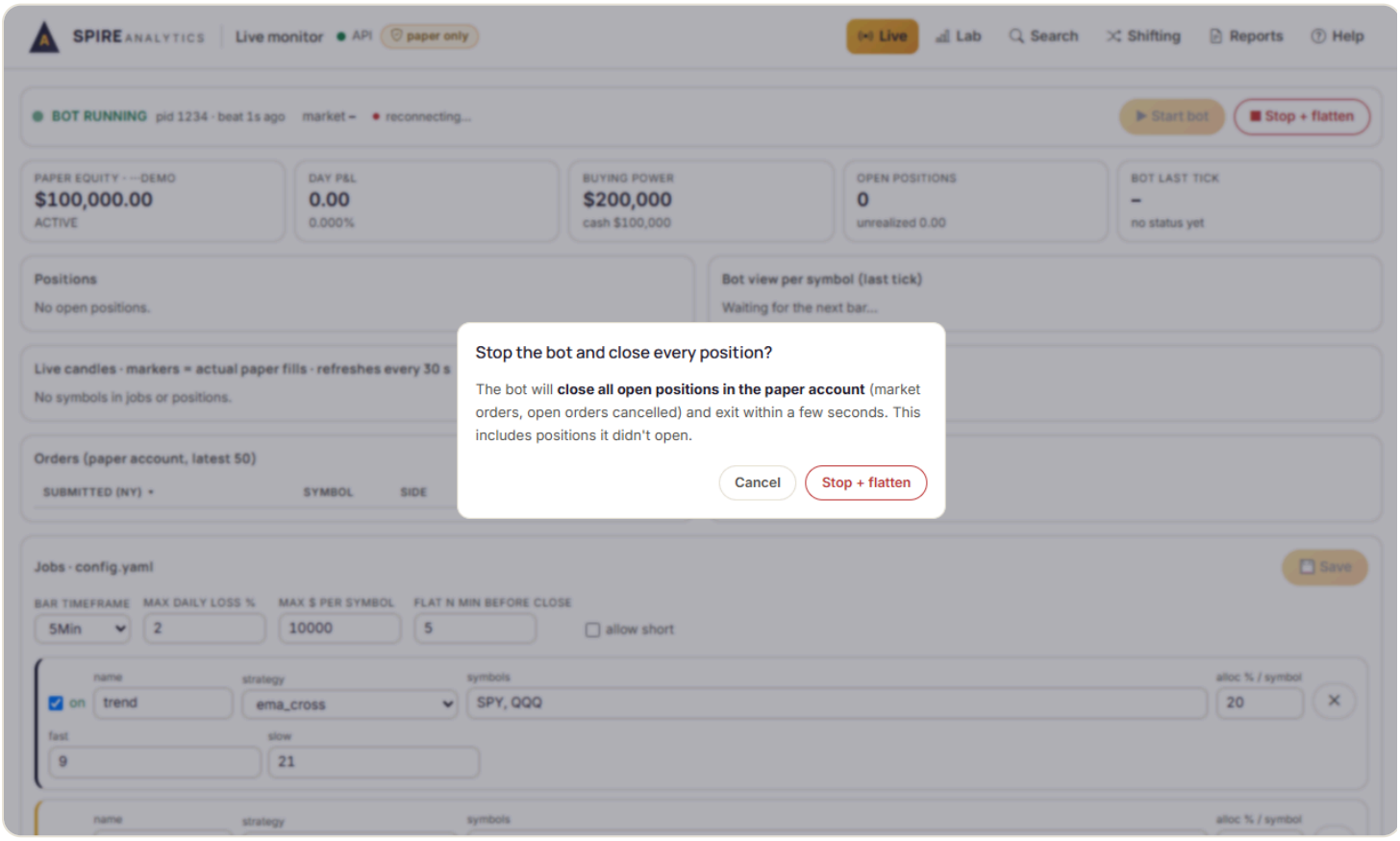


What 'running' looks like: BOT RUNNING, the pid, and a heartbeat a second or two old.

4. Stopping the bot

Normal stop

1. Live page → ■ **Stop + flatten** (or double-click **stop_bot.bat**).
2. Read the confirmation and press **Stop + flatten**. The bot closes **every** open position in the paper account (including ones it didn't open) and exits within a few seconds.
3. Confirm all is flat (see below).



The confirmation dialog. Cancel changes nothing.

Emergency stop

1. If the page doesn't respond: create an empty file named **STOP** (no extension) in E:\autobot. In PowerShell: `New-Item E:\autobot\STOP`. The bot checks for it every few seconds, closes everything and exits; if the bot is stuck, its watchdog does the same within about 10 seconds.
2. If that doesn't work within a minute: on the Alpaca website (paper account) → Positions → **Close all positions**, then end the *python … live.py* process in Task Manager.

Automatic close-of-day flatten

The bot closes all stock positions 5 minutes before the US close (15:55 New York). That is usually **21:55** in Skopje, but it shifts for a few weeks when the US and Europe change their clocks on different dates:

DATES IN 2026	CLOSE-OF-DAY FLATTEN, SKOPJE TIME
01 Jan to 07 Mar	21:55
08 Mar to 28 Mar	20:55
29 Mar to 24 Oct	21:55
25 Oct to 31 Oct	20:55
01 Nov to 31 Dec	21:55

How to confirm all is flat

- Live page → **Positions** says *No open positions*.
- The Alpaca website (paper account) shows no positions and no open orders.
- The status shows **BOT STOPPED** (unless you intend it to keep running).

5. The strategies in plain words

Benchmark

Just hold it `buy_and_hold`

Buy once and keep holding. It is the yardstick every other strategy must beat.

Likes: Markets that rise steadily.

Hates: Long falls: it never steps aside.

Typical behaviour: No trading at all; it simply follows the market up and down.

Trend following

Follow the trend (slow)

sma_cross

Own the stock while its short-term average price is above its long-term average.

Likes: Long, steady trends.

Hates: Choppy, sideways markets where the averages keep crossing.

Typical behaviour: Many small losses while waiting, a few big wins when a trend runs.

Follow the trend

ema_cross

Own the stock while its fast-reacting average price is above its slower average.

Likes: Clear up-trends.

Hates: Sideways markets: it buys late and sells late, again and again.

Typical behaviour: Many small losses, few big wins. Reacts a little faster than the slow version.

Ride the momentum

macd

Own the stock while its upward momentum is strengthening (MACD above its signal line).

Likes: Markets that speed up in one direction.

Hates: Flat, jittery markets.

Typical behaviour: Trades often; small losses in noise, gains when momentum builds.

Follow recent gains

momentum

Own the stock while it is up over the last N bars.

Likes: Persistent trends.

Hates: Sudden reversals.

Typical behaviour: Stays in during trends, gives some back when the trend turns.

Buy the dip / mean reversion

Buy the dip `rsi_reversion`

Buy after a sharp short-term drop (RSI very low) and sell once the price has recovered.

Likes: Calm, sideways markets that bounce back.

Hates: Strong down-trends, where a dip keeps dipping.

Typical behaviour: Many small wins, occasional larger losses.

Buy the stretch `bollinger_reversion`

Buy when the price falls unusually far below its recent average and sell when it returns to the average.

Likes: Calm markets that swing around an average.

Hates: Breakouts and crashes that keep going.

Typical behaviour: Frequent small wins, rare large losses.

Buy the dip in an uptrend `connors_rsi2`

In a long-term uptrend (price above its 200-bar average), buy very short, sharp dips and sell on the bounce.

Likes: Rising markets with brief pullbacks.

Hates: Bear markets (it simply stays out).

Typical behaviour: High win rate, small average gain; designed for daily bars.

Buy the weak close `ibs_reversion`

Buy when a bar closes near its low and sell when a bar closes near its high.

Likes: Index funds in calm markets.

Hates: Trending sell-offs.

Typical behaviour: Short holding times, many small trades; designed for daily bars.

Breakout

Buy the breakout donchian_breakout

Buy when the price makes a new high for the last N bars; sell on a new N-bar low.

Likes: Strong moves that break out of a range.

Hates: Fake breakouts in sideways markets.

Typical behaviour: Low win rate, but the winners can be large.

Intraday (flat every night)

Back to the day's average vwap_reversion

Buy when the price is well below the day's volume-weighted average price and sell when it gets back to it.

Likes: Quiet trading days.

Hates: Days with a strong one-way move.

Typical behaviour: Small, frequent trades within the day.

Trade the opening break opening_range_breakout

Watch the first minutes of the day, then trade in the direction the price breaks out of that range; flat by the close.

Likes: Days with a clear direction early on.

Hates: Days that chop back and forth.

Typical behaviour: One or two trades a day, never held overnight.

First half hour decides the last intraday_momentum

If the market rose in the first half hour, hold it in the last half hour (and the reverse).

Likes: Busy, news-driven days.

Hates: Quiet days with no follow-through.

Typical behaviour: At most one short trade near the close each day.

Break out of the usual noise

noise_area

Only trade when the price moves further from the open than it normally does at that time of day; exit on a trailing stop; flat by the close.

Likes: Strong trending days.

Hates: Range-bound days.

Typical behaviour: Few trades, sits out most of the day.

Strategy shifting

Switch by market mood

regime_switch

Decide whether the market is trending, calm or jumpy, and hand control to the strategy chosen for that mood.

Likes: Markets that stay in one mood for a while.

Hates: Markets that flip mood constantly (switching costs).

Typical behaviour: Behaves like whichever strategy is in charge; stays flat when the market is jumpy if set to.

Back the recent winner

perf_rotation

Every few bars, give control to the strategy that has done best recently.

Likes: Periods where one style keeps working.

Hates: Markets where yesterday's winner is today's loser.

Typical behaviour: Switches now and then; can sit flat when everything has been losing.

Majority vote

ensemble_vote

Ask several strategies and only take the position most of them agree on.

Likes: Clear markets where signals agree.

Hates: Mixed markets, where it mostly stays out.

Typical behaviour: Fewer trades than any single strategy, smoother results.

News-aware

Step aside for news news_guard

Runs another strategy, but takes no new position for a while after an important news story on the stock, and around its earnings releases.

Likes: Markets where news days bring sudden, unpredictable swings.

Hates: Stocks whose best moves happen right after news.

Typical behaviour: Fewer trades on news and earnings days; otherwise identical to the strategy it wraps.

Market moods (regimes)

A strong trend trend

The price is moving steadily in one direction (trend strength above its threshold).

A calm, sideways market range

No strong direction: the price moves back and forth.

A jumpy, high-volatility market high_vol

Price swings are much bigger than usual for this stock.

Not enough history yet warmup

The bot needs more past prices before it can judge the market.

Shifting methods

Switch by market mood
regime_switch

Each bar the bot labels the market trending, sideways or jumpy, then uses the strategy you chose for that label. A new mood must last a few bars before the bot switches, to avoid flip-flopping.

Back the recent winner
perf_rotation

Every K bars the bot checks which of the chosen strategies has done best over the last N bars and hands it control. If all have been losing it can stay out.

Majority vote
ensemble_vote

The bot asks every chosen strategy for its position and follows the majority; with 'average' it takes a partial position in proportion to the votes.

6. What to watch: daily checklist

Look at the Live page a few times a day and read the daily report after the close. ● green = fine, ● amber = keep an eye on it, ● red = act now.

CHECK	GREEN	AMBER	RED: WHAT TO DO
Day P&L vs the daily loss limit	above -1%	-1% to -1.6%	below -1.6% or the bot halted: the limit (2.0%) will stop it; don't restart it the same day
Deepest dip this week (weekly report)	less than 2%	2% to 4%	more than 4%: review the strategies (section 8)
Number of trades vs a normal day	similar to the last 5 reports	twice as many or none at all	many times more: check the log and the jobs; stop the bot if unsure
Errors in the bot log (Live page)	none	an occasional data error (the bot skips that bar)	errors every bar, or CRITICAL/WATCHDOG lines: see troubleshooting
Last tick / heartbeat	last tick within one 5Min bar; beat under 10 s	last tick 1-3 bars old	heartbeat over 90 s, status 'unresponsive', or last tick more than 3 bars old: stop and restart
Strategy switching (reports, Shifting jobs)	0-4 switches a day per symbol	5-8 a day	more than 8 a day: switching costs money; test a larger confirm_bars in the Lab

A symbol stuck in a position	positions change when signals change	an intraday strategy still holding late in the day	a position held with no job on that symbol: close it on the Live page or the Alpaca website

7. Troubleshooting

SYMPTOM	LIKELY CAUSE	WHAT TO DO
Header shows API down	The 'autobot API' window was closed or crashed	Run start_lab.bat again. The bot keeps running on its own; check it on the Live page.
Live page says *reconnecting...*	The API is restarting or not running	Wait 10 seconds; if it stays, run start_lab.bat.
Status unresponsive	The bot process is alive but hasn't beaten for 90 s (stuck)	Press Stop + flatten. If nothing happens within a minute, use the emergency stop. The watchdog also exits a bot stuck for 300 s.
Bot stopped by itself	Daily loss limit, a stall, or the PC slept	Read the bot log tail on the Live page (look for FLATTEN, WATCHDOG, CRITICAL). Restart only if you understand why.
Start refused: 'already running'	A bot is still running	Don't start another. Stop the running one first if needed.
Jobs editor won't save	A rule was broken (message in red), e.g. one symbol in two jobs	Fix what the message says, then Save.
No orders all day	Market closed, signals said 'stay out', or the allocation buys less than 1 share	Check the Bot view per symbol table (signal and target) and the reasons in the daily report.
Report says *reason not recorded*	The trade happened while an older bot version without the decision log was running	Nothing to fix; new trades will have reasons.
No daily report	The lab wasn't running 10 minutes after the close	Reports page → pick the date → Generate.
'data/signal error' lines in the log	Internet or Alpaca data problem	The bot skips those bars. If it lasts, check your connection and Alpaca's status page.

8. Weekly review

1. Reports page → **Last 7 days**. Read the headline, then the charts.

2. Check: the result against simply holding SPY, the number of trades, the worst trade, the deepest dip, and the 'What didn't' list.
3. Compare with the last few weeks' reports; one bad week alone says little.

When to change or switch off a strategy

- It trailed simply holding SPY for three weeks in a row while trading a lot.
- Its deepest dip in a week was more than 4%.
- The report keeps saying 'many small losses in a sideways market' for a trend strategy (or the reverse for a dip-buying strategy).
- It switches strategies more than 8 times a day on a symbol, week after week.

> Always test a change in the Lab first: Lab → run it on Alpaca data (at least 90 days), then Search → Walk-forward. Only change the live job if the walk-forward result is better than what runs today. Change one thing at a time, and note the date so the reports can be compared.

Keeping this document up to date

Run `python scripts/build_sop.py --shots` with the lab running. It re-reads the strategies, limits, jobs and times, takes fresh screenshots and rebuilds this document (Markdown, HTML and PDF).
